

HUJI Research Monitor

Weekly Digest · Week 28 · July 06, 2026

EXECUTIVE SUMMARY

Hebrew University researchers continue to push the boundaries of innovation, with a strong focus on high-impact solutions for unmet medical needs and industrial challenges. This week's digest highlights groundbreaking advancements across cell and gene therapies for autoimmune and neurodegenerative diseases, novel oncology treatments, and precision diagnostics. Furthermore, new technologies in medical devices, advanced materials for infection control and clean energy, and AI-driven drug discovery platforms demonstrate the breadth of our translational research with significant commercial potential.

#1

B cell Maturation Antigen Targeted Chimeric Antigen Receptor T Cell Therapy for Refractory Systemic Lupus Erythematosus and Systemic Sclerosis.

37/50

Main Researcher: Chamutal Gur

[Immunology] [Clinical] [Synthetic Biology]

"Pioneering CAR T-cell therapy offers a transformative approach for severe autoimmune diseases."

WHY NOW

This represents a high-value market shift from oncology to chronic autoimmune conditions, addressing a massive unmet need with a validated therapeutic modality.

COMMERCIAL ANGLE

Developing specialized CAR T-cell therapies for autoimmune indications represents a massive shift from oncology toward high-value chronic disease management markets.

<https://europepmc.org/article/MED/42098533>

#2

Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders.

36/50

Main Researcher: Tamir Ben-Hur

[Neuroscience] [Immunology] [Clinical] [Synthetic Biology]

"Scalable, off-the-shelf stem cell therapy targets myelin repair in neurological disorders like MS."

WHY NOW

This therapy provides a novel regenerative approach for demyelinating diseases, a field with limited options, and offers a scalable cell therapy platform for broad clinical use.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

<https://pubmed.ncbi.nlm.nih.gov/42276062/>

#3

33/50

The effect of a proprietary chlorine-stabilizing surface-coating formulation on healthcare environment disinfection in multidrug-resistant bacteria isolation rooms.

Main Researcher: Oren Zimhony

[Medical Device] [Materials] [Clinical]

"OxiLast™ coating dramatically boosts hospital infection control against superbugs on high-touch surfaces."

WHY NOW

Hospitals face immense pressure to reduce multidrug-resistant infections, making this durable, scalable surface coating an immediate and essential solution.

COMMERCIAL ANGLE

This technology offers a scalable solution for hospitals and long-term care facilities to enhance infection control protocols and reduce the transmission of multidrug-resistant organisms through high-durability antimicrobial coatings.

<https://pubmed.ncbi.nlm.nih.gov/42374536/>

#4

33/50

Recombinant Human Amelogenin Protein Enhances Healing of Osteochondral Injury in a Goat Model.

Main Researcher: Amir Haze

[Medical Device] [Clinical] [Synthetic Biology] [Proteomics]

"Remelix® protein shows promise for significant cartilage and bone repair in joint injuries."

WHY NOW

There's a high demand for effective regenerative treatments for osteoarthritis and traumatic joint injuries, a large and growing market for high-value biologics.

COMMERCIAL ANGLE

Development of a high-value regenerative medicine biologic for treating osteoarthritis and traumatic joint injuries in humans.

<https://pubmed.ncbi.nlm.nih.gov/41925697/>

#5

32/50

APOBEC mutational signatures predict immune checkpoint inhibitor benefit in TMB-low metastatic NSCLC independent of PD-L1 status.

Main Researcher: N Peled

[Diagnostics] [Genomics] [Immunology] [Software/AI]

"Novel diagnostic predicts immunotherapy response in lung cancer, expanding patient access."

WHY NOW

This AI-driven companion diagnostic can identify a new cohort of responders, significantly broadening the market for existing immune checkpoint inhibitors.

COMMERCIAL ANGLE

Development of a companion diagnostic test using machine learning to identify APOBEC signatures in TMB-low lung cancer patients to expand the market for immune checkpoint inhibitors.

<https://pubmed.ncbi.nlm.nih.gov/42378718/>

#6

Targeting primary and metastatic ovarian cancer with a peptide derived from the human NAF-1/CISD2 protein.

29/50

Main Researcher: Rachel Nechushtai

[Drug Discovery] [Clinical] [Immunology]

"A novel peptide targets and inhibits aggressive primary and metastatic ovarian cancer."

WHY NOW

This discovery offers a high-potential, first-in-class therapeutic candidate for a malignancy with urgent unmet needs, attractive for precision oncology pipelines.

COMMERCIAL ANGLE

The peptide represents a high-potential therapeutic candidate for a first-in-class oncology drug, suitable for licensing to pharmaceutical companies focusing on precision cancer therapies or protein-protein interaction inhibitors.

<https://pubmed.ncbi.nlm.nih.gov/42259134/>

#7

Breath-Based Detection of Oral Diseases Using Sensors and Machine Learning.

28/50

Main Researcher: Yael Houri-Haddad

[Medical Device] [Diagnostics] [Software/AI]

"Rapid, non-invasive breathalyzer provides objective diagnostics for various oral diseases."

WHY NOW

This handheld device offers dental clinics a quick, non-subjective screening tool, enhancing early diagnosis and improving patient outcomes in a large market.

COMMERCIAL ANGLE

Development of a non-invasive, rapid-screening handheld breathalyzer for dental clinics to provide objective diagnostics for periodontal health.

<https://pubmed.ncbi.nlm.nih.gov/42379259/>

#8

RadD from *Fusobacterium nucleatum* engages NKp46 to promote antitumor cytotoxicity.

28/50

Main Researcher: Ofer Mandelboim

[Immunology] [Drug Discovery] [Clinical]

"Bacterial protein RadD activates natural killer cells, boosting anti-tumor immunity."

WHY NOW

Modulating NK cell activity is a frontier in cancer immunotherapy, and RadD-based agonists offer a novel approach to enhance tumor killing.

COMMERCIAL ANGLE

Development of RadD-based NK cell agonists or biomimetic immunotherapies to enhance natural killer cell cytotoxicity in oncology.

<https://europepmc.org/article/MED/42065370>

#9

Targeting Temozolomide-Resistant Glioblastoma: Therapeutic Potential of Neuronal Nitric Oxide Synthase Inhibitor.

26/50

Main Researcher: Haitham Amal

[Drug Discovery] [Neuroscience] [Clinical]

"Targeting nitric oxide overcomes drug resistance in deadly glioblastoma, improving chemotherapy."

WHY NOW

This adjuvant therapy extends the clinical utility of standard chemotherapy for glioblastoma, addressing a critical need for resistant brain tumors.

COMMERCIAL ANGLE

Development of a high-value adjuvant therapy to extend the clinical utility of existing standard-of-care chemotherapy for resistant brain tumors.

<https://pubmed.ncbi.nlm.nih.gov/42374654/>

#10

Strategic Template Filtering Accelerates Fragment-Based Peptide Docking.

26/50

Main Researcher: Ora Schueler-Furman

[Drug Discovery] [Software/AI] [Proteomics]

"AI-powered software dramatically accelerates peptide drug design by optimizing binding predictions."

WHY NOW

The demand for efficient AI tools in drug discovery is booming, and this algorithm offers a significant competitive edge for developing peptide therapeutics.

COMMERCIAL ANGLE

Licensing or integrating this high-efficiency docking algorithm into AI-driven drug discovery platforms to accelerate the design of peptide-based therapeutics.

<https://pubmed.ncbi.nlm.nih.gov/42299509/>

#11

IR nanospectroscopy mapping of facet-dependent sulfur poisoning and thermal regeneration on platinum nanocrystals.

26/50

Main Researcher: Elad Gross

[Materials] [Clean Energy]

"Advanced mapping technique paves the way for more durable and efficient industrial catalysts."

WHY NOW

This technology provides a roadmap to engineer next-generation catalysts, critical for improving efficiency and longevity in fuel cells and chemical manufacturing.

COMMERCIAL ANGLE

The ability to map surface-level poisoning provides a roadmap for engineering more durable and efficient catalysts for fuel cells and industrial chemical processes.

<https://pubmed.ncbi.nlm.nih.gov/42065428/>

#12

Safety and Efficacy of Direct Selective Laser Trabeculoplasty in Patients With Ocular Hypertension or Open Angle Glaucoma.

26/50

Main Researcher: David Zadok

[Medical Device] [Clinical]

"Laser treatment offers a safe and effective first-line alternative for glaucoma patients."

WHY NOW

Expanded clinical adoption of this laser procedure addresses patient preference and compliance issues with eye drops, representing a significant market opportunity in ophthalmology.

COMMERCIAL ANGLE

The findings support the commercialization and expanded clinical adoption of laser-based ophthalmic surgical devices as a first-line alternative to traditional glaucoma eye drops.

<https://europepmc.org/article/MED/42148860>